

Tyler LaBonte

Ph.D. Student

Georgia Institute of Technology

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Research Interests

I am interested in foundational aspects of **generalization in machine learning**. My research goal is to advance our scientific and mathematical understanding of deep learning and leverage these fundamental insights to design practical algorithms. I particularly care about ensuring robustness, grounding, and trustworthiness of large models in multimodal and interactive domains. My current focus includes:

- Generalization theory of neural networks and feature learning
- Robustness under distribution shift, task shift, and spurious correlations
- Foundations and applications of large-scale multimodal vision-language models

I also enjoy applying my work to challenging industry-scale research problems. During my PhD, I interned at Google and Microsoft Research (twice), where I most recently worked on development of a 15B multimodal vision-language model with application to computer-use agents.

Education

GEORGIA INSTITUTE OF TECHNOLOGY

2021–2026

Ph.D., Machine Learning

Minor in Mathematics

Advisors: [Vidya Muthukumar](#) and [Jacob Abernethy](#)

UNIVERSITY OF SOUTHERN CALIFORNIA

2017–2021

B.S., Applied and Computational Mathematics, *magna cum laude*

Minor in Computer Science

Advisor: [Shaddin Dughmi](#)

Publications

An asterisk (*) denotes equal contribution. An $(\alpha\text{-}\beta)$ denotes alphabetical author ordering.

PREPRINTS

1. [On the Unreasonable Effectiveness of Last-layer Retraining](#)
John C. Hill, Tyler LaBonte, Xincheng Zhang, and Vidya Muthukumar
Preprint, 2025.

MODEL RELEASES

1. [Phi-4-reasoning-vision and the lessons of training a multimodal reasoning model](#)
Jyoti Aneja, Michael Harrison, Neel Joshi, Tyler LaBonte, John Langford, and Eduardo Salinas ($\alpha\beta$).
Microsoft Research Blog, 2026.

CONFERENCE ARTICLES

1. [Task Shift: From Classification to Regression in Overparameterized Linear Models](#)
Tyler LaBonte*, Kuo-Wei Lai*, and Vidya Muthukumar
AISTATS 2025
INFORMS Applied Probability Society Conference 2025
IMS Workshop on Frontiers of Statistical Machine Learning 2025 (**top-10 award**)
2. [The Group Robustness is in the Details: Revisiting Finetuning under Spurious Correlations](#)
Tyler LaBonte, John C. Hill, Xincheng Zhang, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2024
3. [Towards Last-layer Retraining for Group Robustness with Fewer Annotations](#)
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2023
4. [Scaling Novel Object Detection with Weakly Supervised Detection Transformers](#)
Tyler LaBonte, Yale Song, Xin Wang, Vibhav Vineet, and Neel Joshi
WACV 2023

JOURNAL ARTICLES

1. [Student Misconceptions of Dynamic Programming: A Replication Study](#)
Michael Shindler, Natalia Pinpin, Mia Markovic, Frederick Reiber, Jee Hoon Kim, Giles Pierre Nunez Carlos, Mine Dogucu, Mark Hong, Michael Luu, Brian Anderson, Aaron Cote, Matthew Ferland, Palak Jain, Tyler LaBonte, Leena Mathur, Ryan Moreno, and Ryan Sakuma.
Computer Science Education, 32(3):288–312, 2022
2. [Quantifying the Unknown Impact of Segmentation Uncertainty on Image-Based Simulations](#)
Michael C. Krygier, Tyler LaBonte, Carianne Martinez, Chance Norris, Krish Sharma, Lincoln N. Collins, Partha P. Mukherjee, and Scott A. Roberts
Nature Communications, 12(1):5414, 2021

WORKSHOP ARTICLES

1. [On the Unreasonable Effectiveness of Last-layer Retraining](#)
John C. Hill, Tyler LaBonte, Xincheng Zhang, and Vidya Muthukumar
ICLR 2025 Workshop on Spurious Correlations and Shortcut Learning
2. [Saving a Split for Last-layer Retraining can Improve Group Robustness without Group Annotations](#)
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
ICML 2023 Workshop on Spurious Correlations, Invariance, and Stability
3. [Dropout Disagreement: A Recipe for Group Robustness with Fewer Annotations](#)
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2022 Workshop on Distribution Shifts
4. [Scaling Novel Object Detection with Weakly Supervised Detection Transformers](#)
Tyler LaBonte, Yale Song, Xin Wang, Vibhav Vineet, and Neel Joshi

CVPR 2022 Workshop on Transformers in Vision**THESES**

1. [Finding the Needle in a High-Dimensional Haystack: Oracle Methods for Convex Optimization](#)
Tyler LaBonte
Undergraduate Thesis, University of Southern California, 2021
Winner of the USC Discovery Scholar distinction

MANUSCRIPTS

1. [We Know Where We Don't Know: 3D Bayesian CNNs for Credible Geometric Uncertainty](#)
Tyler LaBonte, Carianne Martinez, and Scott A. Roberts
Manuscript, 2019

Awards

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|--|------|
| 1. IMS Workshop on Frontiers of Statistical Machine Learning Travel Grant (\$500) | 2025 |
| 2. 2 nd Place Research Talk/Poster Presentation – DoD NDSEG CONFERENCE | 2023 |
| 3. Simons Institute Deep Learning Theory Workshop Travel Grant (\$2,000) | 2022 |
| 4. DoD National Defense Science and Engineering Graduate Fellowship (\$170,000) | 2021 |
| • One of two undergraduates to receive both DoD NDSEG and NSF GRFP in Computer Science | |
| 5. NSF Graduate Research Fellowship (\$138,000—declined) | 2021 |
| 6. USC Discovery Scholar (Research distinction for <100 USC graduates) | 2021 |
| 7. Neo Scholar (Top ~100 CS undergraduates in America) – NEO | 2020 |
| 8. U.S.S. Bowfin Memorial Scholarship (\$5,000) | 2020 |
| 9. 1 st Place Computer Vision Project – TREEHACKS, STANFORD UNIVERSITY | 2019 |
| 10. 1 st Place Healthcare AI Project – TREEHACKS, STANFORD UNIVERSITY | 2019 |
| 11. 1 st Place Data Analytics Project – HACKSC, USC | 2019 |
| 12. Admiral Bernard Clarey Memorial Scholarship (\$7,000) | 2018 |
| 13. National Top 20 Ethical Hacking Finalist – MAJOR LEAGUE HACKING | 2018 |
| 14. USC Trustee Scholar (\$250,000) | 2017 |
| 15. USC Viterbi Fellow (\$24,000) | 2017 |
| 16. Dolphin Scholarship (\$13,600) | 2017 |
| 17. Rear Admiral Paul Lacy Memorial Scholarship (\$6,500) | 2017 |
| 18. National Merit Scholar (\$3,000) | 2017 |

Industry Research Experience

1. MICROSOFT RESEARCH Redmond, WA
Machine Learning Research Intern 2025
Advisor: Vibhav Vineet and Neel Joshi
 Joined core development team of a 15B multimodal vision-language model.
2. GOOGLE Sunnyvale, CA
Machine Learning Research Intern 2023
Advisor: Kun Lin
 Developed techniques to leverage Gemini LLM to improve hardware-software code design.
3. MICROSOFT RESEARCH Redmond, WA
Machine Learning Research Intern 2021–2022
Advisor: Vibhav Vineet and Neel Joshi
 Designed Transformer for weakly supervised object detection via multiple instance learning.
4. GOOGLE X Mountain View, CA
Machine Learning Research Intern 2020
Advisor: Daniel R. Silva
 Designed CNN-LSTM architecture for temporal identity preservation in object tracking.
5. SANDIA NATIONAL LABORATORIES Albuquerque, NM
Machine Learning Research Intern 2019–2020
Advisors: Carianne Martinez and Scott A. Roberts
 Developed Bayesian deep learning model for geometric uncertainty in engineering applications.

Talks

1. University of Washington School of Computer Science and Engineering – SEATTLE, WA 2025
 Task Shift: From Classification to Regression in Overparameterized Linear Models
2. Georgia Tech School of Industrial and Systems Engineering – ATLANTA, GA 2024
 Task Shift: From Classification to Regression via Benign Overfitting
3. Georgia Tech Machine Learning Center – ATLANTA, GA 2024
 Task Shift: From Classification to Regression via Benign Overfitting
4. Google DeepMind – MOUNTAIN VIEW, CA 2023
 Towards Last-layer Retraining for Group Robustness with Fewer Annotations
5. Google Cloud Technical Infrastructure – SUNNYVALE, CA 2023
 Large Language Models for Hardware-Software Code Design
6. DoD NDSEG Conference – SAN ANTONIO, TX 2023
 Towards Last-layer Retraining for Group Robustness with Fewer Annotations
7. Microsoft Research – REDMOND, WA 2021
 Weakly Supervised Detection Transformers for Effortless Computer Vision
8. USC Computer Science Theory Group – LOS ANGELES, CA 2021
 The Distance Oracle for Convex Optimization
9. Google X – MOUNTAIN VIEW, CA 2020
 Temporal Identity Preservation in Multiple Object Tracking

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| 10. USC Computer Science Theory Group – LOS ANGELES, CA
3D Bayesian CNNs for Credible Geometric Uncertainty | 2019 |
| 11. USC Center for Artificial Intelligence in Society – LOS ANGELES, CA
3D Bayesian CNNs for Credible Geometric Uncertainty | 2019 |
| 12. USC Center for Artificial Intelligence in Society – LOS ANGELES, CA
Machine Learning Fairness in Word Embeddings | 2019 |

Advising

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| 1. Xincheng Zhang – Georgia Tech MS | 2024–2025 |
| 2. John C. Hill – Georgia Tech BS/MS → Georgia Tech PhD | 2022–2024 |

Teaching

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|---|------|
| 1. Lecturer/Teaching Assistant (8 lectures) Georgia Institute of Technology
CS 7545: Machine Learning Theory | 2024 |
| 2. Lecturer/Teaching Assistant (12 lectures) Georgia Institute of Technology
CS 7545: Machine Learning Theory | 2023 |
| 3. Undergraduate Teaching Assistant University of Southern California
CSCI 270: Introduction to Algorithms and Theory of Computing | 2021 |
| 4. Instructor USC Center for Artificial Intelligence in Society
Introduction to Machine Learning | 2019 |
| 5. Undergraduate Teaching Assistant University of Southern California
CSCI 170: Discrete Methods in Computer Science | 2018 |

Academic Service

Reviewing

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|---|------------|
| 1. CVPR | 2026 |
| 2. ICLR | 2024, 2026 |
| 3. TMLR | 2025, 2026 |
| 4. CVPR Workshop on Demographic Diversity in Computer Vision | 2025 |
| 5. ICLR Workshop on Spurious Correlations and Shortcut Learning | 2025 |
| 6. ICML | 2025 |
| 7. NeurIPS | 2023, 2024 |

Organizing

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| 1. Organizer, Georgia Tech ML Theory Reading Group | 2021–2023, 2025 |
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| 2. System Administrator, Georgia Tech ML Theory GPU Cluster | 2022–2025 |
| 3. Student Organizer, Learning Theory Alliance Mentorship Workshop | 2023 |

Open Source Software

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| 1. Milkshake: Quick and extendable experimentation with classification models
https://github.com/tmlabonte/milkshake | 2023
★ 5 ♪ 3 |
| 2. WS-DETR: Weakly supervised Transformers for scaling novel object detection
https://github.com/tmlabonte/weakly-supervised-detr | 2021–2022
★ 10 ♪ 6 |
| 3. BCNN: 3D Bayesian CNNs for credible geometric uncertainty
https://github.com/sandialabs/bcnn
Transitioned to a production environment by Sandia National Laboratories
19 th most starred Sandia repository out of 693 (March 2025) | 2019–2020
★ 62 ♪ 19 |
| 4. Tendies: Decoupling deep learning development and deployment
https://github.com/tmlabonte/tendies
Transitioned to a production environment by the Air Force Research Laboratory | 2018
★ 37 ♪ 11 |

Other Activities

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| 1. Fleet Captain, Georgia Tech Sailing Club | 2023–2025 |
| 2. House Chair, USC Hawai'i Club | 2020–2021 |
| 3. Vice President of Finance, USC Hawai'i Club | 2019–2020 |